

University of Cape Town

Bayesian Computational Methods

Assignment

Help

Author:
Isabella Lethbridge

Student Number:
LTHISA001

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Introduction

Question a

We can write the probability that $s_i = 3$ as

$$P(t_i) = Pr(T_i > t_i) = \int_{t_i}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{1}{2\sigma^2}(x-\mu)^2)dx.$$

This can be written in terms of Φ after standardising

$$P(t_i) = Pr(\frac{T_i - \mu}{\sigma} > \frac{t_i - \mu}{\sigma}) = 1 - \Phi(\frac{t_i - \mu}{\sigma}).$$

The probability that $s_i = 2$ is denoted as

$$Q(t_i) = Pr(t_i - \tau \leq T_i \leq t_i) = \int_{t_i - \tau}^{t_i} \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{1}{2\sigma^2}(x-\mu)^2)dx.$$

This can be written in terms of Φ after standardising

$$Q(t_i) = Pr(\frac{t_i - \tau - \mu}{\sigma} \leq \frac{T_i - \mu}{\sigma} \leq \frac{t_i - \mu}{\sigma}) = \Phi(\frac{t_i - \mu}{\sigma}) - \Phi(\frac{t_i - \tau - \mu}{\sigma}).$$

The probability that $s_i = 1$ is denoted as

$$R(t_i) = Pr(T_i \leq t_i - \tau) = \int_{-\infty}^{t_i - \tau} \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{1}{2\sigma^2}(x-\mu)^2)dx.$$

This can be written in terms of Φ after standardising

$$R(t_i) = Pr(\frac{T_i - \mu}{\sigma} \leq \frac{t_i - \tau - \mu}{\sigma}) = \Phi(\frac{t_i - \tau - \mu}{\sigma}).$$

We now have

$$L(\boldsymbol{s}, \boldsymbol{t}, \mu, \sigma^2, \tau) = \prod_{i=1}^{n} [1 - \Phi(\frac{t_i - \mu}{\sigma})]^{\mathrm{I}(s_i=3)} \cdot [\Phi(\frac{t_i - \mu}{\sigma}) - \Phi(\frac{t_i - \tau - \mu}{\sigma})]^{\mathrm{I}(s_i=2)} \cdot [\Phi(\frac{t_i - \tau - \mu}{\sigma})]^{\mathrm{I}(s_i=1)}.$$

Question b

Conditional posterior distributions to iteratively sample from using a Gibbs sampler:

$$
\begin{aligned}
T_i \mid \mu, \sigma^2, \tau, s_i = 1, \quad & t_i \sim \mathrm{N}(\mu, \sigma^2) I_{\{T_i \le t_i - \tau\}} \forall i \\
T_i \mid \mu, \sigma^2, \tau, s_i = 2, \quad & t_i \sim \mathrm{N}(\mu, \sigma^2) I_{\{t_i - \tau \le T_i \le t_i\}} \forall i \\
T_i \mid \mu, \sigma^2, \tau, s_i = 3, \quad & t_i \sim \mathrm{N}(\mu, \sigma^2) I_{\{T_i > t_i\}} \forall i \\
\mu \mid \sigma^2, \tau, \boldsymbol{s}, \boldsymbol{t}, \boldsymbol{T} & \sim \mathrm{N}(\bar{T}, \frac{\sigma^2}{n}) I_{\{100 \le \mu \le 160\}} \\
\sigma^2 \mid \mu, \tau, \boldsymbol{s}, \boldsymbol{t}, \boldsymbol{T} & \sim \mathrm{IG}(\frac{n}{2}, \frac{1}{2} \sum_{i=1}^{n} (T_i - \mu)^2) \\
\tau \mid . & \sim \mathrm{Unif}(\tau \in V)
\end{aligned}
$$

where $\mathrm{V} = [50, 180] \cap [l, h]$

Under the Type 2 model, one has observed the moult scores for the birds undergoing moult. Fot those birds the probability density function of the moult score are

$$f_y(y) = \tau f_T(t - y\tau) \qquad 0 < y < 1.$$

After observing the moult scores when $s_i = 2$ we need to update the conditional distributions. Only $T_i \mid \mu, \sigma^2, \tau, s_i = 2, \quad t_i \sim \mathrm{N}(\mu, \sigma^2) I_{\{t_i - \tau \le T_i \le t_i\}} \forall i$ and $\tau \mid . \sim \mathrm{Unif}(\tau \in V)$ change.

We know that $T_i = t_i - y_i \cdot \tau$.

We can write the probability density function (ref eq here) as (for a single obervation)

$$
\begin{aligned}
f_y(y) &= \tau \cdot \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{1}{2\sigma^2}(t - y \cdot \tau - \mu)^2) \\
&= \tau \cdot \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{1}{2\sigma^2}(t^2 - 2ty\tau + (y\tau)^2 - 2\mu t + 2\mu y\tau + \mu^2) \\
&\propto \tau \cdot \exp(-\frac{1}{2\sigma^2}((y\tau)^2 - 2y\tau(t - \mu))) \\
&\propto \tau \cdot \exp(-\frac{y^2}{2\sigma^2}\tau^2 + \frac{y(t - \mu)}{\sigma^2}\tau)
\end{aligned}
$$

The writing it as a likelihood (remember t_i changes for each i)

$$
\begin{aligned}
&\prod_{i=1}^{m} \tau \cdot \exp(-\frac{y_i^2}{2\sigma^2}\tau^2 + \frac{y_i(t_i - \mu)}{\sigma^2}\tau) \\
=&\tau^m \cdot \exp(-(\frac{1}{2\sigma^2}\sum_{i=1}^{m} y_i^2)\tau^2 + (\frac{1}{\sigma^2}\sum_{i=1}^{m} y_i(t_i - \mu))\tau)
\end{aligned}
$$

According to Lemma 5 of Sun et al. (2021)

$$\text{Let } X \sim \mathrm{MHN}(\alpha, \beta, \gamma)$$

a. If $\gamma > 0$ then there exists a random variable V such that $V \mid X \sim \text{Poisson}(\gamma X)$ and $X^2 \mid V \sim \text{Gamma}(\frac{\alpha+V}{2}, \beta)$.

b. If $\gamma < 0$ then there exists a random variable U such that $U \mid X \sim \text{GIG}(\frac{1}{2}, 1, \gamma^2 X^2)$ and $X^2 \mid U \sim \text{Gamma}(\frac{\alpha}{2}, (\beta + \frac{\gamma^2}{U})$.

The PDF for the modified half normal is

$$\begin{aligned} f(x \mid \alpha, \beta, \gamma) &= \frac{2\beta^{\alpha/2} x^{\alpha-1} \exp(-\beta x^2 + \gamma x)}{\Psi[\frac{\alpha}{2}, \frac{\gamma}{\sqrt{\beta}}]} \\ &\propto x^{\alpha-1} \exp(-\beta x^2 + \gamma x) \end{aligned}$$

From (ref eq here) we have $\alpha = m + 1$, $\beta = \frac{1}{2\sigma^2} \sum_{i=1}^{m} y_i^2$ and $\gamma = \frac{1}{\sigma^2} \sum_{i=1}^{m} y_i(t_i - \mu)$.

We can now write the distribution for τ as (Notation is getting a bit confusing here as we now have 2 V's)

$$\tau \sim \text{MHN}(\alpha = m + 1, \beta = \frac{1}{2\sigma^2} \sum_{i=1}^{m} y_i^2, \gamma = \frac{1}{\sigma^2} \sum_{i=1}^{m} y_i(t_i - \mu)) \cdot I_{\{\tau \in \text{V}\}}.$$

From $s_i = 1$ we have

$$\begin{aligned} T_i &\leq t_i - \tau \\ \tau &\leq t_i - T_i \end{aligned}$$

Since t_i changes for every i we have the upper bound $h = \min(t_i - T_i) \forall i$. (I'm not sure if this is the most elegant way of writing this but we can fix it later).

For $s_i = 3$ we have

$$\begin{aligned} t_i - \tau \leq T_i &\leq t_i \\ \tau &\geq t_i - T_i \geq 0 \\ \tau &\geq t_i - (t_i - y_i \tau) \\ \tau &\geq y_i \tau \end{aligned}$$

Since $0 < y_i < 1$ and $\tau > 0$, this is always true.

Thus $\text{V} = [50, min(180, h)]$ and $\tau \in [50^2, \min(100, h)^2]$.

We now consider case (a): $\gamma > 0$

Let m be the number of birds with $s_i = 2$ (the moutling birds).

$$\gamma = \frac{1}{\sigma^2}\sum_{i=1}^{m} y_i(t_i - \mu) > 0$$
$$\sum_{i=1}^{m} y_i(t_i - \mu) > 0$$

$$V \mid \tau \sim \text{Poisson}(\gamma \cdot \tau)$$
$$= \text{Poisson}(\frac{\tau}{\sigma^2}\sum_{i=1}^{m} y_i(t_i - \mu))$$

$$\tau^2 \mid V \sim \text{Gamma}(\frac{m+1+V}{2}, \frac{1}{2\sigma^2}\sum_{i=1}^{m} y_i^2)$$

Next consider case (b): $\gamma < 0$

$$\gamma = \frac{1}{\sigma^2}\sum_{i=1}^{m} y_i(t_i - \mu) < 0$$
$$\sum_{i=1}^{m} y_i(t_i - \mu) < 0$$

$$U \mid \tau \sim \text{GIG}(\frac{1}{2}, 1, (\frac{\tau}{\sigma^2}\sum_{i=1}^{m} y_i(t_i - \mu))^2)$$

$$\tau^2 \mid U \sim \text{Gamma}(\frac{m+1}{2}, (\frac{1}{2\sigma^2}\sum_{i=1}^{m} y_i^2 + \frac{1}{U} \cdot (\frac{1}{\sigma^2}\sum_{i=1}^{m} y_i(t_i - \mu))^2))$$

Now that we have all these lovely distributions we actually need to do this data augmentation shit

We must check the parameterisation of the Gamma dist

Gibbs sampler for the Type 2 model

Let m be the number of birds undergoing moult ($s_i = 2$). The sets $\{i : s_i = 1\}$, $\{i : s_i = 2\}$ and $\{i : s_i = 3\}$ denote the birds that have completed moult, are undergoing moult and have not yet started moult respectively. For the birds undergoing moult the start of moult is not latent anymore since it is fully determined by τ and the observed moult score,

$$T_i = t_i - y_i\tau, \qquad i : s_i = 2.$$

The latent start dates therefore only need to be sampled for the birds with $s_i = 1$ and $s_i = 3$. The conditional distribution of τ is no longer uniform since τ now appears in the density of the observed moult scores. From the derivation above, the conditional of τ is a truncated MHN distribution,

$$\tau \mid \mu, \sigma^2, \boldsymbol{T}, \boldsymbol{y}, \boldsymbol{s}, \boldsymbol{t} \sim \text{MHN}(\alpha, \beta, \gamma)\, I_{\{\tau_L \le \tau \le \tau_U\}}$$

with

$$\alpha = m + 1, \qquad \beta = \frac{1}{2\sigma^2} \sum_{i:s_i=2} y_i^2, \qquad \gamma = \frac{1}{\sigma^2} \sum_{i:s_i=2} y_i (t_i - \mu).$$

The uniform prior on τ and the birds that have completed moult restrict τ to the interval

$$\tau_L = 50, \qquad \tau_U = \min\Big(180, \min_{i:s_i=1} (t_i - T_i)\Big).$$

The birds that have not started moult do not restrict τ since τ does not appear in the condition $T_i > t_i$. The birds undergoing moult also do not restrict τ anymore, since substituting $T_i = t_i - y_i\tau$ into $t_i - \tau \le T_i \le t_i$ gives $y_i\tau \le \tau$ and $y_i\tau \ge 0$, which always holds for $0 < y_i < 1$. This is different from the Type 1 model, where the birds undergoing moult gave the lower bound l.

Sampling from the MHN directly is difficult, so we use data augmentation (Lemma 5 of Sun et al. (2021)). We introduce an auxiliary variable (V if $\gamma > 0$ or U if $\gamma < 0$) and sample it together with τ. The marginal distribution of τ is still the MHN, but each of the two conditionals is a standard distribution that is easy to sample from.

Algorithm

Initialisation. For each chain draw random starting values from the priors, $\mu^{(0)} \sim \text{Uniform}(100, 160)$, $\sigma^{2(0)} \sim \text{InverseGamma}(1, 1)$ and $\tau^{(0)} \sim \text{Uniform}(50, 180)$. Then set $T_i^{(0)} = t_i - y_i\tau^{(0)}$ for all i with $s_i = 2$.

For iteration $k = 1, \ldots, N$:

1. **Birds that have completed moult.** For each i with $s_i = 1$ sample

$$T_i^{(k)} \sim \text{N}\big(\mu^{(k-1)}, \sigma^{2(k-1)}\big)\, I_{\{T_i \le t_i - \tau^{(k-1)}\}}.$$

2. **Birds that have not started moult.** For each i with $s_i = 3$ sample

$$T_i^{(k)} \sim \text{N}\big(\mu^{(k-1)}, \sigma^{2(k-1)}\big)\, I_{\{T_i > t_i\}}.$$

The truncated normal draws in steps 1 and 2 are done using the `RcppTN` package.

3. **Update μ.** Sample

$$\mu^{(k)} \sim \text{N}\Big(\bar{T}, \frac{\sigma^{2(k-1)}}{n}\Big)\, I_{\{100 \le \mu \le 160\}},$$

where $\bar{T}$ is the mean over all n birds, using the sampled $T_i^{(k)}$ for $s_i = 1, 3$ and the calculated $T_i = t_i - y_i\tau^{(k-1)}$ for $s_i = 2$.

4. **Update σ^2.** Sample

$$\sigma^{2(k)} \sim \mathrm{IG}\Big(\frac{n}{2}, \frac{1}{2}\sum_{i=1}^{n}\big(T_i - \mu^{(k)}\big)^2\Big),$$

again using all n start dates.

5. **Update τ (data augmentation).**

   a. Calculate $\alpha = m + 1$, $\beta = \frac{1}{2\sigma^{2(k)}}\sum_{i:s_i=2} y_i^2$ and $\gamma = \frac{1}{\sigma^{2(k)}}\sum_{i:s_i=2} y_i(t_i - \mu^{(k)})$.

   b. Calculate the bounds $\tau_L = 50$ and $\tau_U = \min\big(180, \min_{i:s_i=1}(t_i - T_i^{(k)})\big)$.

   c. If $\gamma > 0$, sample the auxiliary variable and then τ^2:

$$V \mid \tau^{(k-1)} \sim \mathrm{Poisson}(\gamma\,\tau^{(k-1)}), \qquad \tau^2 \mid V \sim \mathrm{Gamma}\Big(\frac{\alpha + V}{2}, \beta\Big)\, I_{\{\tau_L^2 \le \tau^2 \le \tau_U^2\}}.$$

If $\gamma < 0$, sample

$$U \mid \tau^{(k-1)} \sim \mathrm{GIG}(\tfrac{1}{2}, 1, \gamma^2(\tau^{(k-1)})^2), \qquad \tau^2 \mid U \sim \mathrm{Gamma}\Big(\frac{\alpha}{2}, \beta + \frac{\gamma^2}{U}\Big)\, I_{\{\tau_L^2 \le \tau^2 \le \tau_U^2\}}.$$

   d. Set $\tau^{(k)} = \sqrt{\tau^2}$.

   e. Recalculate $T_i = t_i - y_i\tau^{(k)}$ for all i with $s_i = 2$, so that the next iteration uses start dates that agree with the new value of τ.

6. **Store** $\mu^{(k)}$, $\sigma^{2(k)}$ and $\tau^{(k)}$.

The truncated Gamma in step 5c is sampled using the inverse CDF method. Let F be the CDF of the (untruncated) Gamma distribution. We draw $u \sim \mathrm{Uniform}(F(\tau_L^2), F(\tau_U^2))$ and set $\tau^2 = F^{-1}(u)$. Every draw therefore lies in $[\tau_L^2, \tau_U^2]$ and no rejection step is needed. The interval is never empty, since the current τ is at least 50 and the latent T_i for $s_i = 1$ were drawn so that $\tau \le t_i - T_i$.

We run three chains of 50 000 iterations each, every chain with its own random starting values.